Problem Set 2

Section 2.5

28) $UI = \left[\begin{array}{ccc|ccc} 1 & a & b & 1 & 0 & 0 \\ 0 & 1 & c & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$

$\downarrow$

$\left[\begin{array}{ccc|ccc} 1 & 0 & b-ca & 1 & -a & 0 \\ 0 & 1 & c & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$

$\downarrow$

$\left[\begin{array}{ccc|ccc} 1 & 0 & b-ca & 1 & -a & 0 \\ 0 & 1 & 0 & 0 & 1 & -c \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$

$\downarrow$

$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -a & ca-b \\ 0 & 1 & 0 & 0 & 1 & -c \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$

$U^{-1} = \left[\begin{array}{ccc} 1 & -a & ca-b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{array}\right]$

40) $A = \left[\begin{array}{cccc} 1 & -a & 0 & 0 \\ 0 & 1 & -b & 0 \\ 0 & 0 & 1 & -c \\ 0 & 0 & 0 & 1 \end{array}\right]$

nu: EVOLVE

$$[A|I] = \left[\begin{array}{cccc|cccc} 1 & -a & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & -b & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -c & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right]$$

$R_1 = R_1 - (-a)R_2$ ↓

$$\left[\begin{array}{cccc|cccc} 1 & 0 & -ba & 0 & 1 & a & 0 & 0 \\ 0 & 1 & -b & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -c & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right]$$

$R_2 = R_2 - (-b)R_3$ ↓

$$\left[\begin{array}{cccc|cccc} 1 & 0 & -ba & 0 & 1 & a & 0 & 0 \\ 0 & 1 & 0 & -bc & 0 & 1 & b & 0 \\ 0 & 0 & 1 & -c & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right]$$

$R_1 = R_1 - (-ba)R_3$ ↓ $-abc$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & -abc & 1 & a & ba & 0 \\ 0 & 1 & 0 & -bc & 0 & 1 & b & 0 \\ 0 & 0 & 1 & -c & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right]$$

↓

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & a & ba & abc \\ 0 & 1 & 0 & 0 & 0 & 1 & b & bc \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & c \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right]$$

hence $A^{-1} = \begin{bmatrix} 1 & a & ab & abc \\ 0 & 1 & b & bc \\ 0 & 0 & 1 & c \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Section 2.6

13) $E_1 A = \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & b-a & c-a & c-a \\ 0 & b-a & c-a & d-a \end{bmatrix}$

$E_2(E_1 A) = \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & c-b & d-b \end{bmatrix}$

$E_3(E_2 E_1 A) = \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & 0 & d-c \end{bmatrix} = U$

~~$E_4(E_3 E_2 E_1 A)$~~ $= \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-a & \\ 0 & 0 & 0 & \end{bmatrix}$

$L = E_1^{-1} E_2^{-1} E_3^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$

Hence $LU = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & 0 & d-c \end{bmatrix}$

$a \neq b$, $b \neq c$, $c \neq d$, $a \neq 0$

with all factors invertible

18) Suppose $A = LDU$ and $A = L_1 D_1 U_1$. Then $LDU = L_1 D_1 U_1$.

$L_1^{-1} LDU = D_1 U_1$

$L_1^{-1} LD = D_1 U_1 U^{-1}$

LHS is lower triangular, RHS is upper triangular. Since LHS = RHS, that means both sides are diagonal.

Since LHS is diagonal, $L_1^{-1} L$ is diagonal and since RHS is diagonal, $U_1 U^{-1}$ is also diagonal. Since L, L_1, U and U_1 have diagonal 1's, their products also have diagonal 1's. Hence $L_1^{-1} L = I$ and $U_1 U^{-1} = I$, and so $L = L_1$ and $U = U_1$.

Then since $LDU\,ID = D_1 I$ and so $D = D_1$.

26)

$$7K^{-1} = \begin{bmatrix} 6 & 5 & 4 & 3 & 2 & 1 \\ 5 & 10 & 8 & 6 & 4 & 2 \\ 4 & 8 & 12 & 9 & 6 & 3 \\ 3 & 6 & 9 & 12 & 8 & 4 \\ 2 & 4 & 6 & 8 & 10 & 5 \\ 1 & 2 & 3 & 4 & 5 & 6 \end{bmatrix}$$

$$K(7K^{-1}) = \begin{bmatrix} 7 & 0 & 0 & 0 & 0 & 0 \\ 0 & 7 & 0 & 0 & 0 & 0 \\ 0 & 0 & 7 & 0 & 0 & 0 \\ 0 & 0 & 0 & 7 & 0 & 0 \\ 0 & 0 & 0 & 0 & 7 & 0 \\ 0 & 0 & 0 & 0 & 0 & 7 \end{bmatrix}$$

Section 2.7

13) a) $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

b) $\hat{P} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$

36) Lower triangular (L) with 1's on the diagonal, diagonal invertible matrices D, permutation matrices and matrices with $Q^T = Q^{-1}$ are all 'groups'.

Additional matrix groups include upper tirangular matrices U with 1's on the diagonal as well as the set of cP, where c is some scalar and P is a permutation matrix.

40)

a) Let q_i be any column of Q. Then $q_i^T = r_i$, where r_i is a row in Q^T. The dot product of q_i and r_i $(q_i^T q_i)$ gives the entry e_{ii} of $QQ^T = QQ^{-1}$, which is I, and so $e_{ii} = 1$, and since $q_i^T = r_i$, the dot product is $q_{i1}r_{i1} + q_{i2}r_{i2} + \cdots + q_{in}r_{in} = q_{i1}^2 + q_{i2}^2 + q_{i3}^2 + \cdots + q_{in}^2 = e_{ii} = 1$. Applying square root on both sides we get $\sqrt{q_{i1}^2 + q_{i2}^2 + \cdots + q_{in}^2} = 1$ hence $\|q_i\| = 1$ and so $\|q_i\|^2 = 1$

b) Let q_1 and q_2 be any two distinct columns of Q. Then q_1^T is a row in Q^T and hence $q_1^T q_2$ is the entry e_{12} of $Q^TQ = I$ and since this entry is not on the diagonal, it is equal to 0 hence $q_1^T q_2 = 0$

c) $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

Section 3.1

18) a) Let A and B be any symmetric matrices in M. Then:-

$$(A+B)^T = A^T + B^T = A+B$$

So A+B is still symmetric hence statement true.

b) Let A and B be any skew symmetric matrices. Then:-

$$(A+B)^T = A^T + B^T = -A-B = -(A+B)$$

So A+B is skew symmetric hence statement true.

c) The statement is false. Take the unsymmetric matrices $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. Then:-

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 3 & 4 \end{bmatrix}$$

The sum is symmetric.

23) First blank:- Unless b is in the span of the columns of A.

Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$, $b_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ and $b_2 = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$.

Then adding b_1 as a column to A increase the column space ($R^2 \to R^3$) because b_1 is not in the column space of A, but adding b_2 as a column to A changes nothing because $b_2 = a_1 + a_2$ (where a_1 and a_2 are the two columns of A).

~~Ax~~ Since Ax takes linear combinations of ~~A~~ the columns of A, ~~any possb $\in$~~ $Ax = b$ only when b is ~~a~~ a linear combination of the columns of A (and so ~~$\emptyset$~~ a case where adding b as a column of A does not increase the column space).

30) Let s_1, s_2 and t_1, t_2 be any ~~two~~ four vectors in S and t respectively.

Then $s_1 + t_1 \in S+T$ and $s_2 + t_2 \in S+T$.

$(s_1 + t_1) + (s_2 + t_2) = (s_1 + s_2) + (t_1 + t_2)$

$(s_1+s_2)+(t_1+t_2) \in S+T$
and so

Since $(s_1+s_2) \in S$, and $(t_1+t_2) \in T$ due to S and T being subspaces, $S+T$ is closed under addition.

Also, $c(s_1+t_1) = cs_1+ct_1$, where c is any scalar. Since $cs_1 \in S$ and ~~es~~ $ct_1 \in T$, $cs_1+cs_2 \in S+T$ and so $S+T$ is closed under scalar multiplication.

Since $S+T$ is closed under vector addition and scalar multiplication, it is a ~~subs~~vector space.

b) $S+T$ is the set of all linear combinations of vectors in S and vectors in T, assuming S and T are distinct, $S+T$ is a plane, whereas $S \cup T$ is simply those two distinct lines. The span of $S \cup T$ is the set of all vectors that can be ~~reached~~ expressed as a linear combination of $S \cup T$'s vectors, which is by definition $S+T$.

32) Let A and B be any two matrices such that AB is defined

each of

Then ↑ the columns of AB are linear combinations of the columns of A, by ~~and so [A A~~ definition of matrix multiplication, and so [A AB] has the same column space as A since ~~Do~~ appending the columns of AB to A does not add any more ↓ independent vectors linearly

Let $A = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$. Then the column space of A is ~~#~~ $c\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ where $c \in R$, i.e. an infinite line.

Also, $A^2 = AA = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$

$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

~~Whi~~ The column space of A^2 is hence $\{0\}$ which is less than $c(A)$.

Blank:- invertible